

CLASS 9 MATHEMATICS • NCERT 0804CH01

Orienting Yourself: The Use of Coordinates

An exhaustive point-by-point classroom presentation covering every concept, historical root, tactile model, quadrant rule, and the Baudhāyana distance formula directly from the textbook.

 Akash Srivastva (www.akashmaths.online)

Grid-Based Thinking: Roots in Ancient Bharat

Sindhu–Sarasvatī Civilisation

First Systematic Urban Grids: Thousands of years ago, streets were built with striking precision in North–South and East–West directions at uniform intervals (~10 m).

Merchants located warehouses by counting units North–South and East–West from the city centre.

Baudhāyana (c. 800 BCE)

Geometric Lines of Construction: Baudhāyana laid perpendicular East–West and North–South lines to construct sacred altars (Vedi).

Formulated the Baudhāyana–Pythagoras theorem, providing the fundamental basis for modern 2–D coordinate distance.

Ujjayinī: Prime Meridian

Universal Zero–Longitude: Described as early as the 4th century BCE in the Siddhāntas as the central longitude meridian.

Later recorded by Greek geographers (Hipparchus, Ptolemy as 'Ozine') and adopted across world navigation.

The Core Definition of a Coordinate System

A structured framework (like the grid lines on a map or graph sheet) that enables us to use **numbers** to describe the exact physical locations of points or objects in space.

From Celestial Mapping to Cartesian Geometry

★ Āryabhaṭa (c. 499 CE)

Replaced Greek chords with **sines (jya)**. Mapped the skies using *Celestial Coordinates* measured from the ecliptic (path of the Sun).

○ Brahmagupta (c. 628 CE)

Formalised **zero and negative numbers** as algebraic entities. Without Brahmagupta's zero (origin) and negatives, four-quadrant geometry is impossible!

🧭 Al-Bīrūnī & Khayyām

Al-Bīrūnī (c. 1000 CE) perfected the *astrolabe* using Indian methods. Ōmar Khayyām (c. 1100 CE) first solved cubic algebra using coordinates.

X¹ Descartes & Fermat

René Descartes (1637 CE) & Fermat formalised that any point in a 2-D plane is defined by **two numbers**: its distances from two perpendicular axes.

🏛️ The Great Unification: Algebra Meets Geometry

Descartes' coordinate system allows geometric shapes to be represented as algebraic equations, and algebraic equations to be visualized as geometric shapes.

Check Your Understanding: Historical Foundations

- Q1** How did merchants in the Sindhu–Sarasvatī civilisation locate shops or warehouses using grid-based thinking?
- Q2** Why is Brahmagupta's formalisation of zero and negative numbers considered essential for the modern 4-quadrant coordinate plane?
- Q3** Which Indian city was historically designated as the central prime meridian of longitude in ancient astronomical Siddhāntas?
- Q4** What was the major breakthrough achieved by René Descartes and Pierre de Fermat in 1637?

Tactile Mathematics: Spatial Orientation for Reiaan

The Real–World Challenge

- Reiaan is visually impaired; the family has just moved to a new city and a new school.
- His elder sister, Shalini (who just finished Grade 9), applies **Coordinate Geometry** to help him navigate the new space independently.
- She builds a **tactile rectangular grid map** with pins and threads representing objects in the room.

The Tactile Coordinate Scale

- **Chosen Scale:** 1 cm on graph = 1 foot (1 ft) in reality.
- **Pins:** Fixed at precise corner coordinates of the bedroom, bathroom, bed, wardrobe, and doors.
- **Thick Wool:** Connected between pins so Reiaan can feel room boundaries and clearances with his fingertips.

Textbook Inquiry: Why Can't Windows Be Marked on this Map?

A standard floor map is strictly a **2-dimensional (2-D) horizontal projection**. Since windows are positioned at a vertical height above the floor, representing them requires a 3rd dimension (vertical z-axis / elevation)!

Analyzing the Room Layout as a Coordinate System

Room Dimensions & Corners

- **Bedroom Size:** 12 ft × 10 ft. Corners at Origin $O(0, 0)$, $A(12, 0)$, $B(12, 10)$, and $C(0, 10)$.
- **Bed Placement:** $S_1(1, 5)$, $S_2(7, 5)$, $S_3(7, 8)$, $S_4(1, 8)$ → 6 ft long by 3 ft wide.
- **Wardrobe Placement:** $W_1(3, 0)$, $W_2(7, 0)$, $W_3(7, 2)$, $W_4(3, 2)$ → 4 ft wide along x-axis, 2 ft depth.
- **Left Wall:** Acts as the vertical reference (y-axis).
- **Bottom Wall:** Acts as the horizontal reference (x-axis).

Door Coordinates & Accessibility

- **Bedroom Entrance Door (D_1R_1):** Located along the bottom wall between $D_1(8, 0)$ and $R_1(11.5, 0)$.
- **Entrance Width:** $11.5 - 8 = 3.5$ feet (42 inches). Provides comfortable, standard wheelchair clearance!
- **Bathroom Door (B_1B_2):** Located on the y-axis between $B_1(0, 1.5)$ and $B_2(0, 4)$.
- **Bathroom Width:** $4 - 1.5 = 2.5$ feet (30 inches). Narrower than the bedroom entrance door!

Check Your Understanding: Room Layout & Axes

- Q1** How far is the room door D_1R_1 from the left wall (y -axis) of Reiaan's bedroom? How far is it from the x -axis?
- Q2** If D_1 is at $(8, 0)$ and R_1 is at $(11.5, 0)$, what is the door's clear opening width? Can a standard wheelchair pass through easily?
- Q3** The bathroom door has endpoints $B_1(0, 1.5)$ and $B_2(0, 4)$. Is this door wider or narrower than the bedroom entrance door?
- Q4** If the bathroom door is hinged at $B_1(0, 1.5)$ and opens 90° inward into the bedroom, will it hit the wardrobe?

The Structure of the Two-Dimensional Plane

↔ The Horizontal Line: x-axis

The horizontal number line. Distances to the right of Origin O are **positive (+)**; distances to the left are **negative (-)**.

Any point on the x-axis has coordinates of the form: **(x, 0)**.

↑↓ The Vertical Line: y-axis

The vertical number line perpendicular to the x-axis. Distances upward from O are **positive (+)**; downward are **negative (-)**.

Any point on the y-axis has coordinates of the form: **(0, y)**.

⊕ The Origin: Point O

The unique point of intersection of the two perpendicular coordinate axes.

$$0 = (0, 0)$$

REFERENCE ZERO

Both x- and y-coordinates at the origin are precisely 0.

📌 Textbook Convention on Coordinate Notation

While formal mathematics defines a point as $P = (x, y)$, it is customary and convenient on graphs to drop the '=' sign and write simply **P(x, y)**.

Sign Conventions in the Cartesian xy -Plane

Quadrant I

$(+, +)$

$x > 0, y > 0$

Upper Right. Both x and y are positive.

E.g., $A(3, 4), S_3(7, 8)$

Quadrant II

$(-, +)$

$x < 0, y > 0$

Upper Left. Negative x , Positive y .

E.g., $Q(-5, 3), M'(-9, 6)$

Quadrant III

$(-, -)$

$x < 0, y < 0$

Lower Left. Both x and y are negative.

E.g., $M(-3, -4), (-7, -4)$

Quadrant IV

$(+, -)$

$x > 0, y < 0$

Lower Right. Positive x , Negative y .

E.g., $S(3, -5), Z(5, -6)$



Geometric Meaning of Coordinates (x, y)

x -coordinate: Perpendicular distance of point P from the **y -axis**, measured along the x -axis.

y -coordinate: Perpendicular distance of point P from the **x -axis**, measured along the y -axis.

Ordered Pairs: Why Order in (x, y) Matters

Comparing $P(x, y)$ and $Q(y, x)$

- In coordinate geometry, (x, y) is an **ordered pair**. Order determines position!
- Consider $Q(-5, 3)$ vs $S(3, -5)$:
 - $Q(-5, 3)$ lies in **Quadrant II** (5 units left, 3 units up).
 - $S(3, -5)$ lies in **Quadrant IV** (3 units right, 5 units down).
- They represent completely different spatial locations!

The Coincidence Condition

$$(x, y) = (y, x) \Leftrightarrow x = y$$

CONDITION FOR POINTS TO COINCIDE

If $x \neq y$, then $(x, y) \neq (y, x)$. P and Q coincide *only* if $x = y$ (i.e. every point on the line $y = x$, such as $(2, 2)$ or $(-4, -4)$).

Think & Reflect: Axis Coordinates Generalization

1. The x -coordinate of ANY point lying on the y -axis is ALWAYS **0** $\rightarrow (0, y)$.
2. The y -coordinate of ANY point lying on the x -axis is ALWAYS **0** $\rightarrow (x, 0)$.

Check Your Understanding: Quadrants & Ordered Pairs

Q1 In which quadrant do each of the following points lie: A(-3, 5), B(4, -7), C(-6, -2), and D(5, 8)?

Q2 What are the coordinates of a point lying on the y-axis at a distance of 4.5 units below the origin?

Q3 If a point has coordinates (a, b) where $ab > 0$, which quadrants could this point possibly lie in?

Q4 Can the point P(k, -k) ever lie in Quadrant I? What about Quadrant II or IV?

Spatial Problem Solving in Reiaan's House

Study Table Position

Three feet of Reiaan's rectangular study table are at **(8, 9), (11, 9), and (11, 7)**.

- Fourth foot must complete the rectangle at **(8, 7)**.
- Dimensions: Length = 3 ft, Width = 2 ft.
- *Note:* Table height is vertical and cannot be determined from a 2-D floor plan!

Bathroom & Shower Area

Bathroom corners extend along the negative x-axis:

- Corners: O(0, 0), F(0, 10), R(-6, 10), P(-6, 0).
- Shower Area SHWR: S(-6, 6), H(-3, 6), W(-2, 10), R(-6, 10). A non-rectangular quadrilateral with slanted partition HW!

Dining Room & Table

Dining room length extends from P(-6, 0) to A(12, 0) (length = 18 ft) and width = 15 ft extending into negative y.

- Room Corners: (-6, 0), (12, 0), (12, -15), (-6, -15).
- Centre is at (3, -7.5). A 5 ft × 3 ft table centered here has feet at (0.5, -6), (5.5, -6), (5.5, -9), (0.5, -9).

Check Your Understanding: Map Calculations

- Q1** Given the coordinates of three vertices of a rectangle are $(8, 9)$, $(11, 9)$, and $(11, 7)$, what geometric rule confirms the fourth vertex must be $(8, 7)$?
- Q2** Why can we not determine the vertical height of the study table or wardrobe from the floor map?
- Q3** What is the area of Reiaan's bathroom if its vertices are at $O(0, 0)$, $F(0, 10)$, $R(-6, 10)$, and $P(-6, 0)$?
- Q4** If a point lies at $(-6, 0)$, on which coordinate axis does it lie, and in which direction from the origin?

Distance Between Points Parallel to the Axes

↔ Horizontal Segment (Parallel to x-axis)

When two points share the **same y-coordinate**: (x_1, y) and (x_2, y) .

The segment connecting them is strictly horizontal.

$$\text{Distance} = |x_2 - x_1|$$

ABSOLUTE DIFFERENCE OF X-COORDINATES

Example: Between $W_1(3, 0)$ and $W_2(7, 0)$, the distance is $|7 - 3| = 4$ units.

↑↓ Vertical Segment (Parallel to y-axis)

When two points share the **same x-coordinate**: (x, y_1) and (x, y_2) .

The segment connecting them is strictly vertical.

$$\text{Distance} = |y_2 - y_1|$$

ABSOLUTE DIFFERENCE OF Y-COORDINATES

Example: Between $B_1(0, 1.5)$ and $B_2(0, 4)$, the distance is $|4 - 1.5| = 2.5$ units.



What if the segment is NOT parallel to either axis?

When a segment connects points like $A(3, 4)$ and $D(7, 1)$, it is slanted! We cannot simply subtract one coordinate. We must use the **Baudhāyana-Pythagoras Theorem** to form a right triangle.

Derivation via Baudhāyana–Pythagoras Theorem

Step-by-Step Construction (Triangle ADM)

- Let $A = (x_1, y_1)$ and $D = (x_2, y_2)$.
- Construct point $C(x_1, y_2)$ to complete right-angled $\triangle ACD$ with $\angle C = 90^\circ$.
- Base CD:** Horizontal shift = $|x_2 - x_1|$
- Height AC:** Vertical shift = $|y_2 - y_1|$
- By Baudhāyana's Theorem:
 $AD^2 = CD^2 + AC^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

THE UNIVERSAL 2-D DISTANCE FORMULA

$$d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$$

VALID FOR ANY TWO POINTS IN THE CARTESIAN PLANE

Because differences are squared, $(x_2 - x_1)^2 = (x_1 - x_2)^2$. Thus, the order of points does not change the resulting distance!

Reflection in Axes Preserves Distance (Isometry)

↶ Reflection in y-axis

Each point (x, y) maps to $(-x, y)$.

- $A(3, 4) \rightarrow A'(-3, 4)$
- $D(7, 1) \rightarrow D'(-7, 1)$
- $M(9, 6) \rightarrow M'(-9, 6)$

The x-coordinate changes sign; y-coordinate remains identical.

🛡 Distance Invariance

Calculating the lengths of reflected triangle $A'D'M'$:

- $A'D' = \sqrt{4^2 + 3^2} = \mathbf{5 \text{ units}}$
- $D'M' = \sqrt{((-2)^2 + 5^2)} = \mathbf{\sqrt{29} \text{ units}}$
- $M'A' = \sqrt{((-6)^2 + 2^2)} = \mathbf{\sqrt{40} \text{ units}}$

All side lengths are **strictly preserved!** $\triangle ADM \cong \triangle A'D'M'$.

↷ Reflection in x-axis

Each point (x, y) maps to $(x, -y)$.

- $A(3, 4) \rightarrow (3, -4)$
- $D(7, 1) \rightarrow (7, -1)$

Distance is likewise perfectly preserved because $(-y_2 - (-y_1))^2 = (y_2 - y_1)^2$.

💡 Think & Reflect: What Changes and What Remains the Same?

Same: Side lengths, interior angles, and area of the figure remain unchanged.

Changed: Spatial position, quadrant location, and orientation (handedness / clockwise direction) invert!

Check Your Understanding: Distance Calculations

- Q1** What is the distance of any point $P(x, y)$ directly from the Origin $O(0, 0)$?
- Q2** Find the distance between the points $P(-3, 0)$ and $Q(5, 0)$ without using a pencil.
- Q3** If point $A(4, -3)$ is reflected in the x -axis, what are the coordinates of its image A' ? What if it is reflected in the y -axis?
- Q4** Why does changing the order of coordinates in $(x_2 - x_1)^2$ vs $(x_1 - x_2)^2$ make zero difference in distance?

Representative Worked Problems from Exercises

📍 Ex 6: Collinearity Check

Question: Are $M(-3, -4)$, $A(0, 0)$, $G(6, 8)$ on one line without plotting?

Solution via Distance:

- $MA = \sqrt{(0 - (-3))^2 + (0 - (-4))^2} = \sqrt{9 + 16} = 5$
- $AG = \sqrt{(6 - 0)^2 + (8 - 0)^2} = \sqrt{36 + 64} = 10$
- $MG = \sqrt{(6 - (-3))^2 + (8 - (-4))^2} = \sqrt{81 + 144} = \sqrt{225} = 15$

Since $MA + AG = 5 + 10 = 15 = MG$, points M, A, G lie strictly on the same straight line! ✓

✂ Ex 10: Midpoint Formula

Question: Given midpoint $M(-7, 1)$ of segment AB where $A(3, -4)$, find $B(x, y)$.

Deduction: The midpoint coordinates are the averages:

- $x_M = (x_A + x_B)/2 \Rightarrow -7 = (3 + x)/2 \Rightarrow x = -17$
- $y_M = (y_A + y_B)/2 \Rightarrow 1 = (-4 + y)/2 \Rightarrow y = 6$

Coordinates of point B are $(-17, 6)$. ✓

🗨 Ex 12: Points on a Circle

Question: Show $A(1, -8)$, $B(-4, 7)$, $C(-7, -4)$ lie on circle K centered at $O(0, 0)$.

Distance to Center:

- $OA = \sqrt{1^2 + (-8)^2} = \sqrt{1 + 64} = \sqrt{65}$
- $OB = \sqrt{(-4)^2 + 7^2} = \sqrt{16 + 49} = \sqrt{65}$
- $OC = \sqrt{(-7)^2 + (-4)^2} = \sqrt{49 + 16} = \sqrt{65}$

All points are equidistant from O with **Radius** = $\sqrt{65}$ units ≈ 8.06 . They lie on circle K ! ✓

Discussion Challenges for Classroom Exploration

City Street Grid Problem (Problem 14)

Two main roads cross at the city centre (N–S and E–W). Other streets run parallel, 200 m apart.

- Using scale 1 cm = 200 m, intersections are labeled (N–S street, E–W street).
- **Teacher Prompt:** How many intersections can be labeled as (4, 3)? How many as (3, 4)? Why are these two distinct locations?

Computer Graphics Screen (Problem 15)

A screen is 800 pixels wide and 600 pixels high with origin (0, 0) at the bottom-left corner.

- Circle 1: Center A(100, 150), radius = 80 px.
- Circle 2: Center B(250, 230), radius = 100 px.
- **Teacher Prompt:** Does any part of either circle fall off-screen? Do the two circular icons overlap? (Hint: Compare distance AB with $r_1 + r_2$!)

Geometry Check: Proving a Square (Problem 16)

Given vertices A(2, 1), B(–1, 2), C(–2, –1), and D(1, –2). How do students use the distance formula to prove ABCD is a square? (Check that all four sides are equal to $\sqrt{10}$, and both diagonals AC and BD are equal to $\sqrt{20}$!)

Check Your Understanding: Practical Coordinate Systems

Q1 What would a coordinate system look like if negative numbers were never invented? Could it describe the entire 2-D plane?

Q2 If the distance between circle centers is greater than the sum of their radii ($d > r_1 + r_2$), do the circles intersect?

Q3 Point W has x-coordinate equal to 5. Which quadrants can a point H lie in if it is located on a vertical line passing through W?

Q4 How does coordinate geometry allow modern computer screens and smartphones to locate your finger tap?

Complete Concept Checklist & Core Takeaways

✓ Axes, Origin & Coordinates

- ✓ Two perpendicular lines define the 2-D coordinate plane (horizontal x-axis, vertical y-axis).
- ✓ The intersection point is the **Origin $O(0, 0)$** .
- ✓ Point on x-axis: **$(x, 0)$** ; Point on y-axis: **$(0, y)$** .
- ✓ The x-coordinate measures perpendicular distance from the y-axis; y-coordinate from the x-axis.
- ✓ If $x \neq y$, then **$(x, y) \neq (y, x)$** . They coincide if and only if $x = y$.

✓ Quadrants, Distance & Reflection

- ✓ **Four Quadrants:** Q I (+, +), Q II (–, +), Q III (–, –), Q IV (+, –).
- ✓ Horizontal distance = $|x_2 - x_1|$; Vertical distance = $|y_2 - y_1|$.
- ✓ **Baudhāyana Distance Formula:**

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
- ✓ Reflection across axes preserves side lengths and distances (isometry).
- ✓ Midpoint between (x_1, y_1) and $(x_2, y_2) = ((x_1 + x_2)/2, (y_1 + y_2)/2)$.

END OF CHAPTER 1

Geometry and Algebra are but Two Reflections of One Truth.

"With coordinates, the geometer acquires eyes to see algebra, and the algebraist acquires wings to explore space."

✦ Akash Srivastva (www.akashmaths.online)